

From Memory Influence to Source Fidelity: A Causal Audit of Failure-Derived Memory in Embodied Agents

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Preconfirmatory Advisor Draft. This document is a reviewable paper candidate, not a report of unobserved confirmatory outcomes. Existing carrier/qualification evidence is labeled *current evidence*; the decisive closed-loop experiment is labeled *prospective*. The empirical scope is MemoryVLA/LIBERO-Plus unless a second carrier is prospectively opened.

Abstract

Memory-augmented embodied agents increasingly reuse prior experience, yet a behavioral change after memory exposure does not establish that the agent used the source experience faithfully. We study a narrower causal question: when the same embodied history is converted into persistent memory, can we identify whether downstream policy changes are caused by source-faithful information rather than by extra context, serialization, or generic perturbation? Our current evidence establishes an edit-capable memory carrier and motivates an influence–fidelity distinction, but it does not yet authorize a closed-loop causal claim. We therefore define a prospective matched intervention that links a source outcome to a committed memory difference and to a later same-condition replay, with no-op, unrelated-content, and same-content/different-provenance controls. The proposed paper contributes an identification standard for memory influence and source fidelity, together with explicit hard stops that prevent local action sensitivity from being over-interpreted as useful persistent learning.

1 Why this paper exists

A persistent-memory agent may change its action after a retrieved memory is edited. That observation is useful but ambiguous: the policy may react to formatting, added context, a changed token sequence, or generic novelty rather than to the source experience that the memory is supposed to represent. The scientific object of this paper is therefore not “memory helps” and not generic memory management. It is the causal chain

$$\text{source experience} \rightarrow \text{committed memory} \rightarrow \text{later policy use},$$

with an explicit distinction between *influence* (the memory changes behavior) and *fidelity* (the change tracks programmatically verifiable source information).

The practical premise is common to systems that distill successful or failed embodied episodes into reusable memory. If the memory channel can influence later actions without preserving source fidelity, then apparent self-improvement can be driven by a corrupted or semantically ungrounded persistent state. Conversely, if source-faithful changes survive matched controls and affect later behavior, the memory channel becomes a measurable causal interface rather than an opaque prompt augmentation.

2 Closest work and defended residual

MemoryVLA, MemER, and related memory-augmented vision-language-action systems already establish that retrieved experience or history can improve long-horizon control. This paper does not claim novelty for adding memory to a VLA or for showing that history can affect actions. The defended residual is an *identification audit*: hold the carrier and downstream conditions fixed, vary source-grounded information under matched controls, and ask whether the resulting policy change is both causally attributable and source faithful.

This makes the strongest simplification a first-class baseline: if an unrelated edit, no-op rewrite, or same-content provenance perturbation explains the same action change, the source-fidelity interpretation fails even if the headline influence metric is large.

3 Scientific unit and causal object

The confirmatory unit is one frozen tuple consisting of: (i) a baseline source trajectory artifact, (ii) one transient intervention that induces a source-level outcome difference, (iii) the memory derived from that source, and (iv) one downstream same-condition replay from a frozen initial state. Units are selected before treatment outcomes are opened.

For every unit we derive a versioned source-fidelity signature directly from trajectory/state/action records and task/robot kinematic rules. The signature must not use target outcomes or author semantic judgment. The causal object is a branch comparison, not a collection of independently counted provider calls.

4 Current evidence versus open claim

Layer	Current evidence	What remains open
Carrier	The MemoryVLA/LIBERO-Plus path is edit-capable; existing action-shift evidence shows that memory representation can alter policy output.	Carrier sensitivity alone does not identify source-faithful use.
Influence	Prior local influence/fidelity audits motivate separable readouts.	Native closed-loop influence must survive matched no-op/unrelated controls.
Fidelity	A programmatic source signature can be defined without target outcomes.	Downstream change must track source-consistent rather than arbitrary memory differences.
Utility	Not claimed from carrier qualification.	Later same-condition replay must establish the authorized downstream consequence, or the paper remains an audit/diagnostic result.

5 Prospective confirmatory design

5.1 Gate A0: interface and noise qualification

Before confirmatory treatment, the renderer and memory interface must pass a no-op identity test: a serialization-preserving transform T_{noop} should leave the resulting action/trajectory within a prospectively frozen tolerance. Failure at this gate is an interface problem, not evidence against the scientific hypothesis.

5.2 RQ1: influence

Compare the source-grounded memory against the no-op/matched control while freezing task state, carrier, retrieval placement, decoding policy, and evaluation. The endpoint is a native policy/trajectory change on the downstream replay. A positive influence result authorizes only the statement that the persistent channel affected behavior.

5.3 RQ2: content specificity

Apply an unrelated-content edit matched in placement and representation budget. If unrelated edits produce comparable changes, the source-specific interpretation is rejected.

5.4 RQ3: fidelity

Induce a source-level outcome flip that changes the programmatic fidelity signature, derive the corresponding memory, and replay the same downstream condition. The critical comparison asks whether behavior moves in the source-consistent direction. A same-content/different-provenance control tests whether the effect is tied to source information rather than surface content alone.

5.5 RQ4/RQ5: downstream construct validation

Only after influence and fidelity pass do we inspect whether the altered policy use changes native closed-loop utility. Local action changes remain mechanism readouts; they cannot substitute for the primary closed-loop endpoint.

6 Minimum workload and inference

The minimum base confirmatory block is 32 independent source units with four matched conditions (128 downstream runs) plus the no-op qualification where needed. Expansion to 64 units is allowed only by a prospectively stated precision rule, not because early results are unfavorable. Provider/model results are not pooled across identities. Missingness, retries, and evaluator failure rules are frozen before outcome exposure.

7 Baselines that matter

- **No-op edit:** detects serialization/interface sensitivity.
- **Unrelated-content edit:** tests generic perturbation/context effects.
- **Same-content, different provenance:** isolates source metadata from semantic text.
- **Frozen baseline memory:** anchors the incremental effect of the committed change.

A second VLA or memory mechanism is not required for the first carrier-bounded claim. It becomes mandatory only for a generic cross-system measurement claim.

8 Decision rules and hard stops

We stop or narrow the paper if: (1) the no-op gate fails; (2) unrelated edits explain the same influence; (3) the source-fidelity signature cannot be frozen without target-outcome leakage; (4) source-consistent effects fail under same-content controls; or (5) the only robust finding is local action sensitivity with no authorized downstream consequence. These outcomes are informative boundaries, not invitations to add unrelated models or benchmarks.

9 Limitations and scope

The base claim is carrier-bounded to MemoryVLA/LIBERO-Plus. The study does not establish a universal memory law, superiority of a learned memory manager, or multi-carrier generality. The causal intervention intentionally gives the audit controlled access to source-outcome information; the contribution is identification, not a deployment policy for generating that information.

10 Conclusion

The paper asks when a persistent memory channel is not merely behaviorally active but source faithful. Current evidence establishes a usable carrier and a reason to separate influence from fidelity. The decisive causal claim remains prospective and is governed by a matched, outcome-blind intervention with explicit failure conditions. This structure lets a negative result narrow the scientific object cleanly rather than being rescued by additional workload.

Related systems used to define the boundary

MemoryVLA (ICLR 2026) and MemER (ICLR 2026) are treated as prior evidence that memory-augmented embodied policies can use historical information. LIBERO-Plus supplies the primary controlled embodied substrate in the present plan. Exact bibliographic metadata will be normalized before venue submission; the advisor draft uses these systems only to define the scientific boundary above.